Multi-Agent DRL-Based Two-Timescale Resource Allocation for Network Slicing in V2X Communications

Binbin Lu, Yuan Wu, *Senior Member, IEEE*, Liping Qian, *Senior Member, IEEE*, Sheng Zhou, *Senior Member, IEEE*, Haixia Zhang, *Senior Member, IEEE*, and Rongxing Lu, *Fellow, IEEE*

***Abstract*—Network slicing has been envisioned to play a crucial role in supporting various vehicular applications with diverse performance requirements in dynamic Vehicle-to-Everything (V2X) communications systems. However, time-varying Service Level Agreements (SLAs) of slices and fast-changing network topologies in V2X scenarios may introduce new challenges for enabling efficient inter-slice resource provisioning to guarantee the Quality of Service (QoS) while avoiding both resource over-provisioning and under-provisioning. Moreover, the conventional centralized resource allocation schemes requiring global slice information may degrade the data privacy provided by dedicated resource provisioning. To address these challenges, in this paper, we propose a two-timescale resource management mechanism for providing diverse V2X slices with customized resources. In the long timescale, we propose a Proximal Policy Optimization-based multi-agent deep reinforcement learning algorithm for dynamically allocating bandwidth resources to different slices for guaranteeing their SLAs. Under the coordination of agents, each agent only observes its partial state space rather than the global information to adjust the resource requests, which can enhance the privacy protection. Moreover, an expert demonstration mechanism is proposed to guide the action policy for reducing the invalid action exploration and accelerating the convergence of agents. In the short-term time slot, with our proposed Cross Entropy and Successive Convex Approximation algorithm, each slice allocates its available physical resource blocks and optimizes its transmit power to meet the QoS. Simulation results show our proposed two-timescale resource allocation scheme for network slicing can achieve maximum 8.4% performance gains in terms of spectral efficiency while guaranteeing the QoS requirements of users compared to the baseline approaches.**

***Index Terms*—Network slicing, V2X communications, multi-agent deep reinforcement learning, resource allocations.**

Received 20 November 2023; revised 3 June 2024 and 21 August 2024; accepted 28 August 2024. Date of publication 5 September 2024; date of current version 20 December 2024. This work was supported in part by National Natural Science Foundation of China under Grants 62122069, 62072490, and 62071431, in part by the Guangdong Basic and Applied Basic Research Foundation (2022A1515011287), in part by Science and Technology Development Fund of Macau SAR under Grant 0158/2022/A, and in part by MYRG2020-00107-IOTSC. The associate editor coordinating the review of this article and approving it for publication was R. Pasquini. *(Corresponding author: Yuan Wu.)*

Binbin Lu is with the State Key Lab of Internet of Things for Smart City, and the Department of Computer and Information Science, University of Macau, Macau, China.

Yuan Wu is with the State Key Lab of Internet of Things for Smart City, and the Department of Computer and Information Science, University of Macau, Macau, China, and also with the Zhuhai-UM Science & Technology Research Institute, University of Macau, Zhuhai 519072, China (e-mail: yuanwu@um.edu.mo).

Liping Qian is with the College of Information Engineering, Zhejiang University of Technology, Hangzhou 310023, China (e-mail: lpqian@zjut.edu.cn).

Sheng Zhou is with the Beijing National Research Center for Information Science and Technology, Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: sheng.zhou@tsinghua.edu.cn).

Haixia Zhang is with the Shandong Key Laboratory of Wireless Communication Technologies, and the School of Control Science and Engineering, Shandong University, Jinan 250061, China (e-mail: haixia.zhang@sdu.edu.cn).

Rongxing Lu is with the Faculty of Computer Science, University of New Brunswick, Fredericton, NB E3B 5A3, Canada (e-mail: rlu1@unb.ca).

Digital Object Identifier 10.1109/TNSM.2024.3454758

I. Introduction

With the development of the 5th Generation Mobile Networks (5G) and beyond, the Cellular-Vehicle-to-Everything (C-V2X) has provided great technical support for Internet of Vehicles (IoV), which significantly contributes to the success of future Intelligent Transportation Systems (ITS) [1]. With full connections between vehicles, pedestrians and networks, IoV enables various emerging Vehicle-to-Everything (V2X) applications, such as autonomous driving, cooperative collision avoidance and entertainment [2]. The highly diverse Quality of Services (QoS) requirements and limited spectrum resources require more efficient resource allocation mechanism for providing diverse QoS provisioning [3].

As an emerging technology in 5G, network slicing is a promising solution for provisioning diverse QoS services [4]. By creating multiple virtual logical networks on top of common physical networks, network slicing allows network operators to provide customized and dedicated resources for different V2X applications [5]. In this manner, network slicing can satisfy the specific QoS requirements of different V2X applications through offering isolated network resources [6]. However, due to the time-varying Service Level Agreements (SLAs) of slices and fast-changing network topologies, there are still three critical challenges to be addressed in the optimal network slicing for V2X communications.

First, network slicing resource management involves two-level hierarchical problems including the inter-slice resource allocation and the intra-slice resource allocation, which significantly increases the complexity for resource allocation. Though the SLAs prescribe the services, metrics and standards in details, it is challenging to accurately characterize the relationship between the inter-slice bandwidth resource allocation and SLA satisfaction. This problem may lead to

1932-4537 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

bandwidth resource over-provisioning or under-provisioning thereby degrading the resource utilization or not meeting the QoS requirement in the intra-slice resource allocation. Therefore, it is imperative that an efficient resource allocation mechanism should be designed to dynamically determine and adjust the resources assigned to different slices according to time-varying SLAs. Second, the high mobility of vehicles in V2X communications raises more difficulties and uncertainty. The number of vehicles, the network topology and the resource requirements are time-varying in V2X communications [7]. which requires to adjust the inter-slice and intra-slice resource allocation strategies dynamically. Finally, traditional centralized algorithms suffer from data privacy leakages due to requiring to collect the global information. Network slicing can achieve data isolation within slices through using dedicated resources or sharing virtual logical networks. To protect the data privacy, the operators of slices are generally unwilling to expose their sensitive information to others. However, traditional centralized algorithms cannot meet the data privacy isolation requirements of slices due to requiring global information. To tackle these challenges, multi-agent deep reinforcement learning (MADRL) has shown unprecedented performance in network slicing resource management [8]. By continuously interacting with environments, MADRL can not only adapt to dynamic environments and implicitly account for future demands, but also improve the privacy protection with distributed learning framework.

Motivated by the above-mentioned issues, in this paper, we model a two-timescale slice resource allocation problem in V2X communications systems. In the long timescale, the diverse SLAs of slices, the high mobility of vehicles, the computational complexity of resource allocation and the privacy protection feature of slices are considered, and the MADRL algorithm is designed to dynamically determine the approximate inter-slice resource allocation with the objective of maximizing the system spectral efficiency. Then, we accurately model the QoS requirements of the infotainment services (IS) slices and critical message services (CMS) slices, and propose the Cross Entropy and Successive Convex Approximation (CESCA) algorithm to optimize intra-slice physical resource blocks (PRBs) allocation and power control. The main contributions of our paper are summarized as follows.

- We propose a network slicing empowered V2X communications system and model a two-timescale resource allocation problem. In the long timescale, the diverse SLAs of slices, the high mobility of vehicles, the computational complexity of resource allocation and the privacy protection feature of slices are considered, and the MADRL algorithm is designed to dynamically determine the approximate inter-slice resource allocation with the objective of maximizing the weighted sum of average spectral efficiency and energy efficiency.
- For the inter-slice bandwidth resource allocation, a Proximal Policy Optimization (PPO)-based MADRL algorithm is proposed to address the dynamic nature of V2X communications systems and reduce the computational complexity while improving the data privacy protection. To reduce the invalid action exploration and improve the ability to guarantee the QoS requirements, we propose an expert demonstration mechanism to guide the agent to avoid the invalid action decisions.
- For the intra-slice resource allocation, the Cross Entropy (CE) algorithm is proposed for the PRBs allocation optimization. To address the non-convexity of the objective and the delay constraint, Successive Convex Approximation (SCA) algorithm is then proposed for optimizing the power control. With iterative optimization of the PRBs allocation and power control, the energy efficiency of users is maximized while guaranteeing the delay, data rate and reliability requirements.
- We provide extensive simulations to validate the effectiveness and efficiency of our proposed algorithms. The results demonstrate that our proposed MADRL algorithm can achieve approximate optimal performance of centralized DRL algorithm with the global information in terms of the spectral efficiency, and guarantee the vehicles' QoS requirements with the CESCA algorithm.

The remainder of this paper is organized as follows. We review the related studies in Section II. We present the system model and problem formulation in Section III. The decomposition of the problem and our proposed algorithms are shown in Section IV. We demonstrate the performance evaluations in Section V. We finally conclude this work in Section VI.

II. Literature Review

Network slicing involves the logical networks including both the core network and radio access network (RAN). Network slicing in the core network (i.e., core slicing) addresses the optimal virtual network function orchestration, which has been well studied and can be achieved by scaling up computing resources [9], [10], [11], [12]. However, network slicing in RAN (i.e., RAN slicing) is still a challenge due to the scarcity of radio resources, dynamic conditions of wireless channels, and interference [13]. Efficient management of radio resources improves service delivery and utilization of network resources. In general, existing works related to resource allocation for RAN slicing can be grouped into two categories, i.e., the intra-slice resource allocation and the inter-slice resource allocation.

A. Intra-Slice Resource Allocation

Intra-slice resource allocation refers to assigning radio resources to uses for guaranteeing the QoS requirements, which has received significant attention. In [14], Chen et al. proposed an online network slicing scheduling algorithm for optimizing the PRBs allocation and power control in stochastic vehicular networks with time-varying QoS requirements. The objective is to maximize the long-term time-average total system capacity while guaranteeing the stringent reliability and latency requirements and queuing stability constraint. In [15], Zhang et al. proposed an air-ground integrated vehicular network slicing architecture for optimizing multi-dimensional heterogeneous resources including broadcast rate, road side unit (RSU) unicast rate, and vehicle cache size,

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

TABLE I
THE KEY CONTRIBUTIONS OF THE EXISTING WORKS

Existing works	[14]-[19]	[20]	[34]	[33]	[21]-[23]	[32]	[36-37]	[35]	[31]	Our work
Intra-slice	✔	✔	✔	✔				✔	✔	✔
Inter-slice					✔	✔	✔	✔	✔	✔
Diverse SLAs				✔		✔			✔	✔
Intelligent algorithms		✔	✔	✔		✔	✔	✔	✔	✔
Privacy protection			✔				✔			✔

which can save about 40% of the RSU transmission rate while maintaining same QoS requirements. In [16], Hossain et al. utilized numerology-based time division duplex RAN slicing to enhance the system capability, in which the duplex ratio, numerology scheme, bandwidth and power allocation are jointly optimized. Aiming to enable more flexibility and openness, Motalleb et al. in [17] proposed an open RAN system based on network slicing and optimized the PRBs and power allocation to promote balance between the heterogeneous services subject to end-to-end latency constraints.

To improve radio utilization, researchers have recently resorted to multiplexing eMBB and URLLC slices in RAN networks. In [18], Setayesh et al. proposed the penalized SCA algorithm for the PRBs and transmit power allocation optimization to guaranteeing the isolation between the enhanced Mobile Broadband (eMBB) and Ultra-Reliable and Low-Latency Communications (URLLC) slices. Considering multiplexing eMBB and URLLC slices, Yang et al. in [19] re-cut the PRBs and formulated the URLLC and eMBB multiplexing problem as a two-timescale resource allocation optimization problem, in which the long-term total slice utility is maximized subject to the transmission power and network bandwidth constraints. In [20], Alsenwi et al. proposed a DRL algorithm to optimize the resource allocation for eMBB and URLLC coexistence slicing RAN networks.

B. Inter-Slice Resource Allocation

The radio resources of RAN should be allocated to each slice before its operation according to the SLA. There have been several works addressing the inter-slice resource allocation problem with uncertain and diverse requirements. In [21], Wang et al. proposed a conservative strategy for resource allocation to guarantee QoS requirements of services under the worst-case peak traffic conditions, which results in expensive cost and low resource efficiency. In [22], Fendt et al. proposed an adjustable safety factor for slice resource provisioning with a probabilistic guarantee, in which each slice suffers a certain risk of network overload. To address this issue, Luu et al. in [23] proposed a prioritized admission control mechanism based on a resource reservation scheme, in which the number of slices is maximized while reservation considering the uncertainties in slice resource requests. Besides, several existing works have discussed optimizing the slice admission control for inter-slice resource allocation, such as [24], [25], [26], [27]. However, the aforementioned works fail to characterize the relationship between the resource requests and SLAs of slices. Furthermore, traditional optimization methods have suboptimal performance in large, complex and dynamic V2X networks.

Recently, DRL has been considered as a promising solution for resource allocation in RAN networks. DRL has the benefits of both policy-based methods and auction-based method while reducing computational complexity in dynamic environments [28], [29], [30]. In [31], Abdellatif et al. proposed the PPO algorithm to select the optimal node and links for slices and allocate their communication and computation resources. In [32], Alcaraz et al. proposed a model-free reinforcement learning algorithm with a self-assessment mechanism to assign each slice sufficient resources to meet its SLA. The SLA violation rate is kept under an acceptable level but still approximates 30 percent during online learning process. In [33], Hua et al. applied the generative adversarial network-powered DRL to enable request-aware resource allocation for diversified services. The system utility involving spectral efficiency and satisfaction ratio is maximized while guaranteeing the queuing delay and transmission delay constraints. In [34], Mei et al. proposed a DRL algorithm comprising the long short term memory mechanism and actor-critic mechanism to provide customized network slices for diverse V2V services. In [35], Sun et al. proposed a DRL algorithm to dynamically allocate distinct bandwidth resources to slices for supporting the D2D-based V2V communications. In [36], to further improve the efficiency and security of the decentralized algorithm, Boateng et al. proposed a novel hierarchical framework for guiding the underloaded slices to resold spectrum resources to overloaded slices, in which multi-agent DRL algorithm is designed to find the optimal pricing and demand prediction strategies. In [37], Tang et al. proposed a distributed digital twin-assisted DRL algorithm to optimize the computing, storage and bandwidth resources allocations for each node and link of slices. However, the works [31], [32], [33], [34], [35] require to collect global information to guide the action policy, which influences the nature of data isolation and privacy protection provided by network slicing. Although adopting the MADRL algorithm, [36], [37] pays limited attention to optimizing both the inter-slice and intra-slice resource allocation.

The key contributions of existing works and the gaps compared to our work are presented in Table I. In this work, we propose a two-timescale resource allocation framework to enable efficient both intra-slice and inter-slice resource managements in V2X communications. We design the MADRL algorithm not only adapts to dynamic SLAs efficiently but also

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

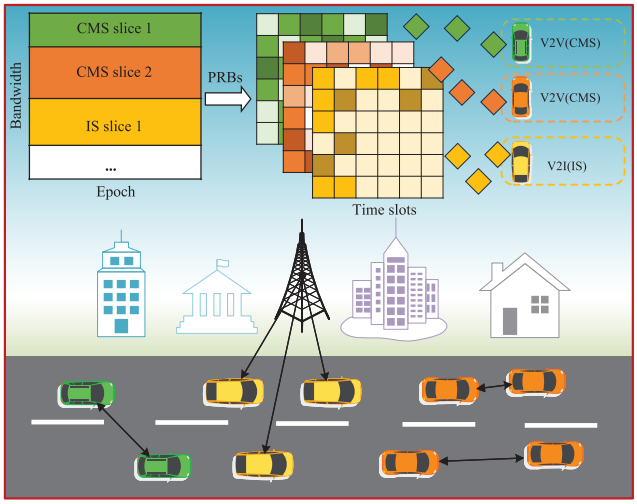


Fig. 1. The framework of our proposed network slicing empowered V2X communications systems.

enhances privacy protection by avoiding the use of global slice information.

III. System Model and Problem Formulation

A. System Model

As shown in Figure 1, we consider a typical V2X communications system consisting of a base station (BS) and several ground vehicles, similar to the scenario described in [14], where the V2X communications can be divided into two types: vehicle to infrastructure (V2I) communications and vehicle to vehicle (V2V) communications. The high-capacity V2I communications can support various IS services such as video streaming and crowdsensing. Moreover, safety-critical messages with ultra-reliable and low-latency requirements in CMS services should be forwarded to nearby vehicles via V2V communications [38]. To this end, in this paper, M IS slices and N CMS slices are constructed to enable V2I communications and V2V communications, which are denoted by $\mathcal{M} = \{1, 2, \ldots, M\}$ and $\mathcal{N} = \{1, 2, \ldots, N\}$, respectively.

We consider, as [38], [39], that the continuous time is divided into K decision epochs. We denote $\mathcal{K} = \{0, 1, \ldots, K\}$ as the set of epochs. Let $\mathcal{I}_m^{s,k}$ and $\mathcal{I}_n^{c,k}$ denote the sets of vehicles in the m-th IS slice and the n-th CMS slice, respectively. We assume that the sets of vehicles in slices are fixed at each epoch. However they might vary between different epochs. At the beginning of each epoch, the operators of slices who provide services to vehicles should submit the bandwidth request according to their SLAs. To meet the required SLAs and keep slice isolation strictly, the bandwidth assigned to each slice is used exclusively, i.e., the bandwidth resources are orthogonal between different slices. The allocation principle depends on the type of the slice and its SLA, which will be detailed in Section IV. The bandwidth assigned to each slice is fixed in the period of decision epoch but can be scaled up or down in following epochs. We denote $W_m^s(k)$ and $W_n^c(k)$ as the bandwidth assigned to the m-th IS slice and the n-th CMS slice at the k-th epoch, respectively. Let W denote the total bandwidth. The total bandwidth assigned to all slices needs to satisfy

$$\sum_{m\in\mathcal{M}} W_m^s(k) + \sum_{n\in\mathcal{N}} W_n^c(k) \le W, \quad \forall k \in \mathcal{K}. \tag{1}$$

Each epoch is then divided into T time slots, and the duration of a slot is τ. We use $\mathcal{T} = \{0, 1, \ldots, T\}$ to denote the set of time slots. At each time slot, slices need to assign their own bandwidth resources to vehicles to meet the QoS requirements. We consider that the Orthogonal Frequency Division Multiple Access (OFDMA) technology is adopted to serve vehicles in slices. The number of physical resource blocks (PRBs) in each slice depends on the assigned bandwidth, which can be expressed as

$$L_m^s(k) = W_m^s(k)Z, \quad \forall m \in \mathcal{M}, k \in \mathcal{K}, \tag{2}$$
$$L_n^c(k) = W_n^c(k)Z, \quad \forall n \in \mathcal{N}, k \in \mathcal{K}, \tag{3}$$

where Z is the number of PRBs per unit bandwidth. Let $\mathcal{L}_m^{s,k}$ and $\mathcal{L}_n^{c,k}$ denote the sets of PRBs in IS and CMS slices, respectively.

B. IS Slices

The bandwidth resources assigned to IS slices can enable high-capacity V2I communications to support various entertainment and traffic-related V2X applications. In IS slices, to improve the bandwidth utilization and the transmission rate of V2I communications, we assume that each PRB can be allocated to multiple vehicles and each vehicle can occupy multiple PRBs simultaneously. And the powerful inter-cell interference cancelation techniques are used to reduce the inter-user interference. Let the binary variables $x_{l,i}^{m,k}(t) \in \{0, 1\}$ indicate whether the l-th PRB is assigned to the i-th vehicle in the m-th IS slice at the t-th time slot of the k-th epoch. If the i-th vehicle uses the l-th PRB, $x_{l,i}^{m,k}(t) = 1$, otherwise $x_{l,i}^{m,k}(t) = 0$. Therefore, the instantaneous signal-to interference-plus-noise-ratio (SINR) of the i-th vehicle can be calculated by

$$\gamma_{l,i}^{m,k}(t) = \frac{p_{l,i}^{m,k}(t) h_{l,i}^{m,k}(t)}{\sum\limits_{i'\in\mathcal{I}_m^{s,k}, i'\neq i} x_{l,i'}^{m,k}(t) p_{l,i'}^{m,k}(t) h_{l,i'}^{m,k}(t) + \sigma^2}, \tag{4}$$

where $p_{l,i}^{m,k}(t)$ denotes the instantaneous transmit power when the i-th vehicle uses the l-th PRB, and σ^2 denotes the noise power of the Additive White Gaussian Noise (AWGN). $h_{l,i}^{m,k}(t)$ indicates the channel gains. The instantaneous transmission rate of the i-th vehicle can be given by

$$r_i^{m,k}(t) = \sum_{l\in\mathcal{L}_m^{s,k}} x_{l,i}^{m,k}(t) B \log_2\left(1 + \gamma_{l,i}^{m,k}(t)\right), \tag{5}$$

where B is the bandwidth per PRB.

A minimum instantaneous transmission rate is required to ensure smooth service experience of the IS slices, which can be expressed as

$$r_i^{m,k}(t) \ge r_m^{s,\min}, \quad \forall i \in \mathcal{I}_m^{s,k}, m \in \mathcal{M}, k \in \mathcal{K}, \tag{6}$$

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

where $r_m^{s,\min}$ denotes the minimum instantaneous transmission rate requirement in the *m*-th IS slice.

Let $A_i^{m,k}(t)$ denote the data arrival rate of the *i*-th vehicle in the *m*-th IS slice at the *t*-th time slot of the *k*-th epoch, which follows the Poisson process with parameter $\lambda_i^{m,k}$. Hence, at the beginning of the $(t+1)$-th time slot, the queue length of the *i*-th vehicle is given by

$$Q_i^{m,k}(t+1) = \max\Big\{ Q_i^{m,k}(t) + \tau A_i^{m,k}(t) - \tau r_i^{m,k}(t), 0 \Big\}. \quad (7)$$

In particular, $Q_i^{m,k}(0)$ indicates the queue length of the *i*-th vehicle at the beginning of the *k*-th epoch. We assume that $Q_i^{m,k}(0)$ is the average queue length across all vehicles over the whole $(k-1)$-th epoch, which can be given by

$$Q_i^{m,k}(0) = \frac{1}{T}\frac{1}{I_m^{s,k-1}} \sum_{i \in \mathcal{I}_m^{s,k-1}} \sum_{t \in \mathcal{T}} Q_i^{m,k-1}(t). \quad (8)$$

The queue length is linear to the average queuing delay, which dominates the delay over a V2I link according to [40]. Let $Q_m^{s,\max}$ denote the tolerable queue length. The delay constraint of the *i*-th vehicle in the *m*-th IS slice with tolerable probability threshold $P_m^{s,\text{d-max}}$ can be given by

$$\Pr\Big\{ Q_i^{m,k}(t) \geq Q_m^{s,\max} \Big\} \leq P_m^{s,\text{d-max}}, \quad \forall i \in \mathcal{I}_m^{s,k}, m \in \mathcal{M}, k \in \mathcal{K}. \quad (9)$$

C. CMS Slices

The CMS slices are mainly constructed to support safety-critical messages services based on V2V communications, in which ultra-reliable and low-latency requirements should be satisfied. To guarantee the stringent reliability requirement on the signal-to-nosie-ratio (SNR), we assume that each vehicle is allocated dedicated PRBs, i.e., each PRB can be allocated to only one vehicle. Compared with V2I links in IS slices, there is no inter-user interference between V2V links, which can effectively enhance the reliability of safety-critical messages transmission. We denote the PRBs allocation decisions by binary variables $x_{l,i}^{n,k} \in \{0,1\}$. Hence, $x_{l,i}^{n,k}$ should satisfy

$$\sum_{i \in \mathcal{I}_n^{c,k}} x_{l,i}^{n,k}(t) \leq 1, \quad \forall l \in \mathcal{L}_n^{c,k}, n \in \mathcal{N}, t \in \mathcal{T}, k \in \mathcal{K}. \quad (10)$$

The instantaneous signal-to-nosie-ratio (SNR) can be given by

$$\gamma_i^{n,k}(t) = \frac{p_{l,i}^{n,k}(t) h_{l,i}^{n,k}(t)}{\sigma^2}, \quad (11)$$

where $p_{l,i}^{n,k}(t)$ and $h_{l,i}^{n,k}(t)$ indicate the instantaneous transmit power and channel gains, respectively. Since there is no inter-user interference, the SNR of vehicles in CMS slices should only depend on the instantaneous transmit power and channel gains, but not be affected by the assigned PRBs. To this end, we assume that the instantaneous transmit power of each vehicle is equal when occupying different PRBs, i.e., $p_{l,i}^{n,k}(t) = p_{l',i}^{n,k}(t), \forall l, l' \in \mathcal{L}_n^{c,k}$.

The V2V communications in CMS slices typically use short packet transmission, in which the instantaneous transmission rate with certain codeword decoding error probability ε cannot be precisely represented by Shannon's capacity. The finite blocklength channel coding regime is considered as an alternative for such short packet scenarios according to [41]. In finite blocklength channel coding regime, the instantaneous transmission rate of the *i*-th vehicle in the *n*-th CMS slice at the *t*-th time slot of the *k*-th epoch can be given by

$$r_i^{n,k}(t) = \sum_{l \in \mathcal{L}_n^{c,k}} x_{l,i}^{n,k}(t) B \left[\log_2\Big(1+\gamma_i^{n,k}(t)\Big) - \sqrt{\frac{V_i^{n,k}(t)}{C}} Q^{-1}(\varepsilon) \log e \right], \quad (12)$$

where C denotes the length of codeword block, $Q^{-1}(\cdot)$ indicates the inverse of the Gaussian Q-function, and $V_i^{n,k}(t)$ is the channel dispersion, given by

$$V_i^{n,k}(t) = 1 - \frac{1}{\Big(1+\gamma_i^{n,k}(t)\Big)^2}. \quad (13)$$

With respect to the high reliable requirement, similar to [42], the outage probability is considered as a significant evaluating index to measure the reliability of V2V communications. To ensure the transmission reliability, the probability that the instantaneous SNR of any V2V communication in the *n*-th CMS slice is less than the minimum instantaneous SNR $\gamma_n^{c,\min}$ should be below the tolerable threshold $P_n^{c,\text{r-max}}$, which can be expressed as

$$\Pr\Big\{ \gamma_i^{n,k}(t) \leq \gamma_n^{c,\min} \Big\} \leq P_n^{c,\text{r-max}}, \quad \forall i \in \mathcal{I}_n^{c,k}, n \in \mathcal{N}, k \in \mathcal{K}. \quad (14)$$

The $A_i^{n,k}(t)$ denotes the data arrival rate of the *i*-th vehicle in the *n*-th CMS slice at the *t*-th time slot of the *k*-th epoch, which obeys Poisson process with $\lambda_i^{n,k}$. The queue length of the *i*-th vehicle at the beginning of the $(t+1)$-th time slot can be expressed by

$$Q_i^{n,k}(t+1) = \max\Big\{ Q_i^{n,k}(t) + \tau A_i^{n,k}(t) - \tau r_i^{n,k}(t), 0 \Big\}, \quad (15)$$

$$Q_i^{n,k}(0) = \frac{1}{T}\frac{1}{I_n^{c,k}} \sum_{i \in \mathcal{I}_n^{c,k}} \sum_{t \in \mathcal{T}} Q_i^{n,k-1}(t). \quad (16)$$

Hence, the delay constraint of the *i*-th vehicle in the *n*-th CMS slice subject to the tolerable probability threshold $P_n^{c,\text{d-max}}$ can be given by

$$\Pr\Big\{ Q_i^{n,k}(t) \geq Q_n^{c,\max} \Big\} \leq P_n^{c,\text{d-max}}, \quad \forall i \in \mathcal{I}_n^{c,k}, n \in \mathcal{N}, k \in \mathcal{K}, \quad (17)$$

where $Q_n^{c,\max}$ and $P_n^{c,\text{d-max}}$ are the tolerable queue length and probability threshold of the *n*-th CMS slice, respectively.

D. Problem Formulation

In this paper, we divide the time into the epoch and time slot two dimensions. At the beginning of each epoch, the bandwidth resources assigned to each slice should be optimized to improve the spectral efficiency of the V2X

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

communications systems. Let $\mathbf{W} = \{W_m^s(k), W_n^c(k), \forall m, n\}$ be the bandwidth allocation variables matrix. Therefore, in the epoch dimension, we aim to maximize the time-average system spectral efficiency $\overline{SE}$ and the optimization variables are the assigned bandwidth of all slices (i.e., $\mathbf{W}$), which can be given by

$$\overline{SE} = \lim_{K\to\infty} \frac{1}{K} \sum_{k\in\mathcal{K}} \frac{\overline{SC^{\text{slot}}}}{\sum_{m\in\mathcal{M}} W_m^s(k) + \sum_{n\in\mathcal{N}} W_n^c(k)}, \quad (18)$$

where $\overline{SC^{\text{slot}}}$ is the average system capacity of each time slot, and it can be expressed as

$$\overline{SC^{\text{slot}}} = \frac{1}{T} \sum_{t\in\mathcal{T}} \left[\sum_{m\in\mathcal{M}} \sum_{i\in\mathcal{I}_m^{s,k}} r_i^{m,k}(t) + \sum_{n\in\mathcal{N}} \sum_{i\in\mathcal{I}_n^{c,k}} r_i^{n,k}(t) \right]. \quad (19)$$

During each epoch, given the bandwidth assignment strategy, we aim to maximize the average system energy efficiency while guaranteeing the delay and reliability requirements of the IS and CMS slices. The optimization variables are the PRBs assigned to vehicles (i.e., $x_{l,i}^{m,k}(t)$ and $x_{l,i}^{n,k}(t)$) and transmit power (i.e., $p_{l,i}^{m,k}(t)$ and $p_{l,i}^{n,k}(t)$) at each time slot. Let $\mathbf{X} = \{x_{l,i}^{m,k}(t), x_{l,i}^{n,k}(t), \forall l, i, m, n\}$ and $\mathbf{P} = \{p_{l,i}^{m,k}(t), p_{l,i}^{n,k}(t), \forall l, i, m, n\}$ denote the PRBs allocation variables matrix and vehicle transmit power matrix, respectively. The average system energy efficiency can be expressed by

$$\overline{EE} = \lim_{K\to\infty} \frac{1}{K} \sum_{k\in\mathcal{K}} \overline{EE^{\text{slot}}}, \quad (20)$$

where $\overline{EE^{\text{slot}}}$ is the average system energy efficiency of each time slot, and its value is given by

$$\overline{EE^{\text{slot}}} = \frac{1}{T} \sum_{t\in\mathcal{T}} \left[\sum_{m\in\mathcal{M}} \frac{\sum_{i\in\mathcal{I}_m^{s,k}} r_i^{m,k}(t)}{\sum_{i\in\mathcal{I}_m^{s,k}} \sum_{l\in\mathcal{L}_m^{s,k}} \left(x_{l,i}^{m,k} p_{l,i}^{m,k}(t) + p_c \right)} + \sum_{n\in\mathcal{N}} \frac{\sum_{i\in\mathcal{I}_n^{c,k}} r_i^{n,k}(t)}{\sum_{i\in\mathcal{I}_n^{c,k}} \sum_{l\in\mathcal{L}_n^{c,k}} \left(x_{l,i}^{n,k} p_{l,i}^{n,k}(t) + p_c \right)} \right]. \quad (21)$$

In particular, p_c is the internal circuit power consumption of vehicles.

Therefore, the optimization problem is formulated as follows

$$(\text{SEEE}) \quad \max_{\{\mathbf{W},\mathbf{X},\mathbf{P}\}} \mu\overline{SE} + \overline{EE} \quad (22)$$

$$\text{s.t.} \sum_{m\in\mathcal{M}} W_m^s(k) + \sum_{n\in\mathcal{N}} W_n^c(k) \le W, \quad \forall k \in \mathcal{K}, \quad (22a)$$

$$\sum_{i\in\mathcal{I}_n^c} x_{l,i}^{n,k}(t) \le 1, \quad \forall l \in \mathcal{L}_n^c(k), n \in \mathcal{N}, t \in \mathcal{T}, k \in \mathcal{K}, \quad (22b)$$

$$p_{l,i}^{m,k}(t) \le p_m^{s,\max}, \quad \forall l \in \mathcal{L}_m^{s,k}, i \in \mathcal{I}_m^{s,k}, m \in \mathcal{M}, t \in \mathcal{T}, k \in \mathcal{K}, \quad (22c)$$

$$p_{l,i}^{n,k}(t) \le p_n^{c,\max}, \quad \forall l \in \mathcal{L}_n^{c,k}, i \in \mathcal{I}_n^{c,k}, n \in \mathcal{N}, t \in \mathcal{T}, k \in \mathcal{K}, \quad (22d)$$

$$r_i^{m,k}(t) \ge r_m^{s,\min}, \quad \forall i \in \mathcal{I}_m^{s,k}, m \in \mathcal{M}, t \in \mathcal{T}, k \in \mathcal{K}, \quad (22e)$$

$$\Pr\left\{ Q_i^{m,k}(t) \ge Q_m^{s,\max} \right\} \le P_m^{s,\text{d-max}}, \quad \forall i \in \mathcal{I}_m^{s,k}, m \in \mathcal{M}, t \in \mathcal{T}, k \in \mathcal{K}, \quad (22f)$$

$$\Pr\left\{ \gamma_{n,i}^{c,k}(t) \le \gamma_n^{c,\min} \right\} \le P_n^{c,\text{r-max}}, \quad \forall i \in \mathcal{I}_n^{c,k}, n \in \mathcal{N}, t \in \mathcal{T}, k \in \mathcal{K}, \quad (22g)$$

$$\Pr\left\{ Q_i^{n,k}(t) \ge Q_n^{c,\max} \right\} \le P_n^{c,\text{d-max}}, \quad \forall i \in \mathcal{I}_n^{c,k}, n \in \mathcal{N}, t \in \mathcal{T}, k \in \mathcal{K}. \quad (22h)$$

Problem (SEEE) is a strictly mixed integer nonlinear programming (MINLP) problem, which is rather challenging to solve in general. Problem (SEEE) is also a two-time scaled optimization problem and a non stationary process, which greatly increases the problem complexity. Besides, in spite of fixing the PRBs allocation matrix $\mathbf{X}$, the objective function is still non-convex with respect to the instantaneous transmit power allocation matrix $\mathbf{P}$. Moreover, the constraints (22f), (22g) and (22h) are time dependent, the optimal PRBs allocation and transmit power control decisions need to be continuously learned to adapt to the varying network conditions.

To address the difficulties, we can vertically decompose Problem (SEEE) into a top-problem and a bottom-problem according to the timescale as follows. In the bottom-problem, given the bandwidth assigned to slices at each epoch, i.e., $\mathbf{W}$, we jointly optimize the intra-slice PRBs allocation and power control to maximize the average system energy efficiency at each time slot. In the top-problem, we further optimize the optimal bandwidth assigned to different slices to maximize the weighted sum of spectral efficiency and energy efficiency from a long-term perspective. The details are presented as follows.

1) Bottom-Problem: Given the bandwidth allocation matrix $\mathbf{W}$, we can decompose Problem (SEEE) into $M + N$ bottom-problems, which can be divided into two types: *Resource Allocation for IS Slices* (IS-BOT) and *Resource Allocation for CMS Slices* (CMS-BOT).

- *Resource Allocation for IS Slices* (IS-BOT): The objective function of Problem (IS-BOT) is to maximize the average energy efficiency of each IS slice, i.e.,

$$\overline{EE^s} = \frac{1}{T} \sum_{t\in\mathcal{T}} \frac{\sum_{i\in\mathcal{I}_m^{s,k}} r_i^{m,k}(t)}{\sum_{i\in\mathcal{I}_m^{s,k}} \sum_{l\in\mathcal{L}_m^{s,k}} \left(x_{l,i}^{m,k} p_{l,i}^{m,k}(t) + p_c \right)}. \quad (23)$$

  (IS-BOT): $\max \overline{EE^s}$
  Subject to: Constraints (22c), (22e) and (22f),
  Variables: $x_{l,i}^{m,k}(t), p_{l,i}^{m,k}(t)$.

- *Resource Allocation for CMS Slices* (CMS-BOT): The objective function of Problem (CMS-BOT) is to

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

maximize the average energy efficiency of each CMS slice, i.e.,

$$\overline{EE^c} = \frac{1}{T}\sum_{t\in\mathcal{T}} \frac{\sum\limits_{i\in\mathcal{I}_n^{c,k}} r_i^{n,k}(t)}{\sum\limits_{i\in\mathcal{I}_n^{c,k}}\sum\limits_{l\in\mathcal{L}_n^{c,k}}\left(x_{l,i}^{n,k}p_{l,i}^{n,k}(t)+p_c\right)}. \quad (24)$$

(CMS-BOT): $\max \overline{EE^c}$

Subject to: Constraints (22b), (22d), (22g), and (22h),

Variables: $x_{l,i}^{n,k}(t), p_{l,i}^{n,k}(t)$.

2) Top-Problem: To find the optimal bandwidth assigned to different slices, Problem (TOP-Bandwidth) is written as

(TOP-Bandwidth): $\max \mu\overline{SE} + \overline{EE}$

Subject to: Constraint (22a),

Variables: $W_m^s(k), W_n^c(k)$.

IV. Problem Transformation and Proposed Algorithms

In this section, we will present Problem (IS-BOT) and Problem (CMS-BOT) decomposed from the bottom-problem and their corresponding solutions in details. Then, we exploit the MADRL algorithm based on PPO algorithm with expert demonstration mechanism to dynamically optimize the inter-slice bandwidth assignment in the top-problem. Finally, the complexity analysis of our proposed algorithms is provided.

A. Resource Allocation for IS Slices (IS-BOT)

Problem (IS-BOT) is still a MINLP problem. To address the difficulty, we propose a layered algorithm, where Problem (IS-BOT) is decomposed into two subproblems, i.e., Problem (IS-BOT-PRB) and Problem (IS-BOT-POW). To solve Problem (IS-BOT-PRB), we propose the CE-Algorithm to find out the optimal PRBs allocation strategy. In Problem (IS-BOT-POW), given the value of $x_{l,i}^{m,k}(t)$ at each time slot, we further optimize the power control variables $p_{l,i}^{m,k}(t)$.

1) Solutions for Problem (IS-BOT-POW): Given the value of $x_{l,i}^{m,k}(t)$, we can transform Problem (IS-BOT) as follows

(IS-BOT-POW): $\max \overline{EE\text{slot}}$

Subject to: Constraints (22c), (22e)and (22f),

Variables: $p_{l,i}^{m,k}(t)$.

The constraint (22f) is a global constraint, which is determined by the decisions over all time slots. To overcome this difficulty, we transform the probabilistic constraint on delay to a deterministic constraint. According to [43], a non-negative random variables X should meet the principle as follows

$$\Pr\{X \geq a\} \leq \frac{E\{X\}}{a}. \quad (25)$$

Due that the queue length is always a non-negative variables in the whole Markov process, the delay

$$\Pr\left\{Q_i^{m,k}(t) \geq Q_m^{s,\max}\right\} \leq \frac{E\left[Q_i^{m,k}(t)\right]}{Q_m^{s,\max}} \leq P_m^{s,\text{d-max}}$$

$$\Rightarrow \quad r_i^{m,k}(t) \geq \lambda_i^{m,k} - \frac{Q_m^{s,\max}P_m^{s,\text{d-max}}}{\tau}. \quad (26)$$

Let $r_i^{m,k,\text{req}} = \max\{r_m^{s,\min}, \lambda_i^{m,k} - \frac{Q_m^{s,\max}P_m^{s,\text{d-max}}}{\tau}\}$. The constraints (22e) and (26) can be combined as

$$r_i^{m,k}(t) \geq r_i^{m,k,\text{req}}. \quad (27)$$

The constraint (27) is still non-convex, and we can rewrite it as follows

$$\begin{aligned} r_i^{m,k}(t) &= \sum_{l\in\mathcal{L}_m^{s,k}} x_{l,i}^{m,k}(t)B\log_2\left(1+\gamma_{l,i}^{m,k}(t)\right) \\ &= \sum_{l\in\mathcal{L}_m^{s,k}} B\log_2\left(1+\frac{x_{l,i}^{m,k}(t)p_{l,i}^{m,k}(t)h_{l,i}^{m,k}(t)}{\sum\limits_{i'\neq i}x_{l,i'}^{m,k}(t)p_{l,i'}^{m,k}(t)h_{l,i'}^{m,k}(t)+\sigma^2}\right) \\ &= \sum_{l\in\mathcal{L}_m^{s,k}} B\log_2\left(\sum_{i\in\mathcal{I}_m^{s,k}} x_{l,i}^{m,k}(t)p_{l,i}^{m,k}(t)h_{l,i}^{m,k}(t)+\sigma^2\right) \\ &\quad - \sum_{l\in\mathcal{L}_m^{s,k}} B\log_2\left(\sum_{i'\neq i} x_{l,i'}^{m,k}(t)p_{l,i'}^{m,k}(t)h_{l,i'}^{m,k}(t)+\sigma^2\right) \\ &\geq r_i^{m,k,\text{req}}. \end{aligned} \quad (28)$$

To tackle the non-convexity of the second term of $r_i^{m,k}(t)$, we adopt the operations of Successive Convex Approximation (SCA). Specifically, we define

$$r_i^{m,k,\text{loss}}(t) \triangleq -\sum_{l\in\mathcal{L}_m^{s,k}} B\log_2\left(\sum_{i'\neq i} x_{l,i'}^{m,k}(t)p_{l,i'}^{m,k}(t)h_{l,i'}^{m,k}(t)+\sigma^2\right). \quad (29)$$

Then, we transform $r_i^{m,k,\text{loss}}(t)$ in terms of any given local point $p_{l,i'}^{m,k,\text{loc}}(t)$ by the Taylor Expansion Formula and set the first-order Taylor Expansion as its lower bound, i.e.,

$$\begin{aligned} r_i^{m,k,\text{loss}}(t) &\geq r_i^{m,k,\text{lb}}(t) \\ &\triangleq -\sum_{l\in\mathcal{L}_m^{s,k}} B\log_2\left(\sum_{i'\neq i} x_{l,i'}^{m,k}(t)p_{l,i'}^{m,k,\text{loc}}(t)h_{l,i'}^{m,k}(t)+\sigma^2\right) \\ &\quad - \sum_{l\in\mathcal{L}_m^{s,k}} \frac{B}{\ln 2}\frac{\sum\limits_{i'\neq i} x_{l,i'}^{m,k}(t)h_{l,i'}^{m,k}(t)\left[p_{l,i'}^{m,k}(t)-p_{l,i'}^{m,k,\text{loc}}(t)\right]}{\sum\limits_{i'\neq i} x_{l,i'}^{m,k}(t)p_{l,i'}^{m,k,\text{loc}}(t)h_{l,i'}^{m,k}(t)+\sigma^2}. \end{aligned} \quad (30)$$

To this end, the constraint (27) is transformed into a convex constraint, which can be written as

$$\begin{aligned} r_i^{m,k}(t) &\geq \sum_{l\in\mathcal{L}_m^{s,k}} B\log_2\left(\sum_{i\in\mathcal{I}_m^{s,k}} x_{l,i}^{m,k}(t)p_{l,i}^{m,k}(t)h_{l,i}^{m,k}(t)+\sigma^2\right) \\ &\quad - \sum_{l\in\mathcal{L}_m^{s,k}} B\log_2\left(\sum_{i'\neq i} x_{l,i'}^{m,k}(t)p_{l,i'}^{m,k,\text{loc}}(t)h_{l,i'}^{m,k}(t)+\sigma^2\right) \\ &\quad - \sum_{l\in\mathcal{L}_m^{s,k}} \frac{B}{\ln 2}\frac{\sum\limits_{i'\neq i} x_{l,i'}^{m,k}(t)h_{l,i'}^{m,k}(t)\left[p_{l,i'}^{m,k}(t)-p_{l,i'}^{m,k,\text{loc}}(t)\right]}{\sum\limits_{i'\neq i} x_{l,i'}^{m,k}(t)p_{l,i'}^{m,k,\text{loc}}(t)h_{l,i'}^{m,k}(t)+\sigma^2} \\ &\geq r_i^{m,k,\text{req}} \end{aligned} \quad (31)$$

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 SCA Algorithm to Solve Problem (IS-BOT-POW)

Initialization: Initialize $p_{l,i}^{m,k,\mathrm{loc}}(t)$ and $\rho^{m,k,\mathrm{loc}}(t)$, and set a small yet positive constant ϵ^{thr} as the end gap.
while $\epsilon^{\mathrm{upd}} > \epsilon^{\mathrm{thr}}$ **do**
  Calculate $p_{l,i}^{m,k}(t)$ and $\rho^{m,k}(t)$ by optimizing Problem (IS-BOT-POW-C).
  Update $\epsilon^{\mathrm{upd}} = |\rho^{m,k,\mathrm{loc}}(t) - \rho^{m,k}(t)|$.
  Update $p_{l,i}^{m,k,\mathrm{loc}}(t) = p_{l,i}^{m,k}(t)$ and $\rho^{m,k,\mathrm{loc}}(t) = \rho^{m,k}(t)$.
end while
Output: P^*.

To make the objective function convex, we introduce a slack variable $\rho^{m,k}(t)$ and get the equivalent result as follows

$$\overline{EE^s} = \frac{1}{T}\sum_{t\in\mathcal{T}} \rho^{m,k}(t), \tag{32a}$$

$$\sum_{i\in\mathcal{I}_m^{s,k}} r_i^{m,k}(t) - \rho^{m,k}(t) \sum_{i\in\mathcal{I}_m^{s,k}} \sum_{l\in\mathcal{L}_m^{s,k}} \left(x_{l,i}^{m,k} p_{l,i}^{m,k}(t) + p_c\right) \geq 0. \tag{32b}$$

In summary, Problem (IS-BOT-POW) can be transformed into a convex optimization problem as follows

$$\begin{aligned} \text{(IS-BOT-POW-C):} \quad & \max \frac{1}{T}\sum_{t\in\mathcal{T}} \rho^{m,k}(t) \\ \text{Subject to:} \quad & \text{Constraints (22c), (31)and (32b),} \\ \text{Variables:} \quad & p_{l,i}^{m,k}(t), \rho^{m,k}(t). \end{aligned}$$

The detailed process of our proposed SCA algorithm is summarized in **Algorithm 1**.

2) Solutions for Problem (IS-BOT-PRB): The PRBs allocation strategy will influence the intra-user interference. Thus, in Problem (IS-BOT-PRB), we further optimize the PRBs allocation strategy to maximize the system energy efficiency while guaranteeing the constraints (22c), (22e) and (22f).

$$\begin{aligned} \text{(IS-BOT-PRB):} \quad & \max \overline{EE^s} \\ \text{Subject to:} \quad & \text{Constraints (22c) (22e)and (22f),} \\ \text{Variables:} \quad & x_{l,i}^{m,k}(t). \end{aligned}$$

The CE method is adopted to optimize the probability of PRBs decision variables $x_{l,i}^{m,k}(t)$, which is detailed in **Algorithm 2**. In particular, we assume that the binary variables $x_{l,i}^{m,k}(t)$ obeys the Bernoulli distribution with the parameter $\theta_{l,i}^{m,k}(t)$ as follows

$$\Phi_{l,i}^{k,t}\left(x_{l,i}^{m,k}(t)\right) = \theta_{l,i}^{m,k}(t)^{x_{l,i}^{m,k}(t)} \left(1-\theta_{l,i}^{m,k}(t)\right)^{\left(1-x_{l,i}^{m,k}(t)\right)}. \tag{33}$$

Then, we iteratively update $\Phi_{l,i}^{k,t}(x_{l,i}^{m,k}(t))$ until the optimal solution is found out to maximize $\overline{EE^s}$. The iterative process consists of the following phases

- Generate $S^{\max}$ random binary samples of $\mathbf{X}$ with the current $\Phi_{l,i}^{k,t}(x_{l,i}^{m,k}(t))$.

Algorithm 2 CE Algorithm to Solve Problem (IS-BOT-PRB)

Initialization: Initialize $\theta_{l,i}^{m,k}(t) = 0.5, \forall l \in \mathcal{L}_m^{s,k}, \forall i \in \mathcal{I}_m^{s,k}$, and set a small yet positive constant ϵ^{thr} as the end gap.
while $\epsilon^{\mathrm{upd}} > \epsilon^{\mathrm{thr}}$ **do**
  Generate $S^{\max}$ random binary samples of $\mathbf{X}$ with the current $\Phi_{l,i}^{k,t}(x_{l,i}^{m,k}(t))$.
  Calculate $\overline{EE^s}$ by solving Problem (IS-BOT-POW-C) with given $\mathbf{X}$.
  if any of the constraints (22c), (22e) and (22f) is not satisfied **then**
    Set $\overline{EE^s} = 0$.
  end if
  Sort the samples in descending order by $\overline{EE^s}$.
  Select the first S^{sam} samples.
  Calculate $\theta_{l,i}^{m,k,*}(t)$ with eq. (34).
  Calculate $\theta_{l,i}^{m,k,\mathrm{upd}}(t)$ with eq. (35).
  Update $\epsilon^{\mathrm{upd}} = \max_{l\in\mathcal{L}_m^{s,k}, i\in\mathcal{I}_m^{s,k}} |\theta_{l,i}^{m,k,\mathrm{upd}}(t) - \theta_{l,i}^{m,k}(t)|$.
  Set $\theta_{l,i}^{m,k}(t) = \theta_{l,i}^{m,k,\mathrm{upd}}(t)$, and update $\Phi_{l,i}^{k,t}(x_{l,i}^{m,k}(t))$.
end while
Output: $\mathbf{X}^*$.

- Given $x_{l,i}^{m,k}(t)$, calculate $\overline{EE^s}$ by solving Problem (IS-BOT-POW-C).
- Let $\overline{EE^s} = 0$, if any of the constraints (22c), (22e) and (22f) is not satisfied.
- Select S^{sam} samples in the order of the values of $\overline{EE^s}$, and calculate $\theta_{l,i}^{m,k,*}(t)$ based on these samples according to the function as follows

$$\theta_{l,i}^{m,k,*}(t) = \arg \max_{0\leq\theta_{l,i}^{m,k}(t)\leq 1} \mathbb{E}\left[\Phi_{l,i}^{k,t}\left(x_{l,i}^{m,k}(t)\right)\right]. \tag{34}$$

- Update $\theta_{l,i}^{m,k,\mathrm{upd}}(t)$ by

$$\theta_{l,i}^{m,k,\mathrm{upd}}(t) = (1-\varrho)\theta_{l,i}^{m,k}(t) + \varrho\theta_{l,i}^{m,k,*}(t), \tag{35}$$

  where ϱ denotes the weight factor.
- Update $\Phi_{l,i}^{k,t}(x_{l,i}^{m,k}(t))$ with $\theta_{l,i}^{m,k,\mathrm{upd}}(t)$.
- Generate $S^{\max}$ improved samples in the next iteration.

B. Resource Allocation for CMS Slices (CMS-BOT)

To make Problem (CMS-BOT) more tractable, we firstly deal with the probabilistic constraint (22g) on reliability. We assume that the $h_{l,i}^{n,k}(t)$ is independent and identically distributed (i.i.d) exponential random variables with unit mean [44]. We can derive the feasible power region of the outage probability as

$$\begin{aligned} \Pr\left\{\gamma_{n,i}^{c,k}(t) \leq \gamma_n^{c,\min}\right\} &= \Pr\left\{\frac{p_{l,i}^{n,k}(t) h_{l,i}^{n,k}(t)}{\sigma^2} \leq \gamma_n^{c,\min}\right\} \\ &= \int_0^{\frac{\gamma_n^{c,\min}\sigma^2}{p_{l,i}^{n,k}(t)}} e^{-h_{l,i}^{n,k}(t)}\, dh_{l,i}^{n}(t) \\ &= 1 - \exp\left(-\frac{\gamma_n^{c,\min}\sigma^2}{p_{l,i}^{n,k}(t)}\right) \leq P_n^{c,\text{r-max}}. \end{aligned} \tag{36}$$

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

Then, we can get the equivalent result of inequality (36) as follows

$$p_{l,i}^{n,k}(t) \geq \frac{\gamma_n^{c,\min}\sigma^2}{\ln\left(\frac{1}{1-P_n^{c,\text{r-max}}}\right)}. \tag{37}$$

Similar to the constraint (22f), we can convert the constraint (22h) as

$$r_i^{n,k}(t) \geq r_i^{n,k,\text{req}}, \tag{38}$$

where $r_i^{n,k,\text{req}} = \lambda_i^{n,k} - \frac{Q_n^{c,\max} P_n^{c,\text{d-max}}}{\tau}$.

Since the transmission rate of vehicles in CMS slices is represented by the finite blocklength channel coding, the constraint (38) is not convex. To address the difficulty, we introduce the SNR threshold $\gamma^{c,\text{thr}} = 7\,\text{dB}$ according to [45]. If $\gamma_i^{n,k}(t) \geq \gamma^{c,\text{thr}}$, the effect of channel dispersion will be negligible, such that channel dispersion $V_i^{n,k}(t)$ can be set as 1. By adding the SNR threshold, the constraint (38) can be further transformed into two convex constraints as follows

$$r_i^{n,k}(t) = \sum_{l\in\mathcal{L}_n^{c,k}} x_{l,i}^{n,k}(t) B \left[\log_2\left(1+\gamma_i^{n,k}(t)\right) - \sqrt{\frac{1}{C}} Q^{-1}(\varepsilon)\log e \right], \tag{39}$$

$$\gamma_i^{n,k}(t) \geq \gamma^{c,\text{thr}}. \tag{40}$$

Similar to Problem (IS-BOT), we introduce a slack variable $\rho^{n,k}(t)$ to make the objective function of Problem (CMS-BOT) convex, which can be expressed as

$$\overline{EE^c} = \frac{1}{T}\sum_{t\in\mathcal{T}} \rho^{n,k}(t), \tag{41a}$$

$$\sum_{i\in\mathcal{I}_n^{c,k}} r_i^{n,k}(t) \geq \rho^{n,k}(t) \sum_{i\in\mathcal{I}_n^{c,k}} \sum_{l\in\mathcal{L}_n^{c,k}} \left(x_{l,i}^{n,k}(t) p_{l,i}^{n,k}(t) + p_c \right). \tag{41b}$$

To this end, given the $x_{l,i}^{n,k}(t)$, we can transform Problem (CMS-BOT) into a convex problem as follows

$$\begin{aligned} \text{(CMS-BOT-POW-C):} \quad & \max \frac{1}{T}\sum_{t\in\mathcal{T}} \rho^{n,k}(t) \\ \text{Subject to:} \quad & \text{Constraints (37), (39), (40) and (41b),} \\ \text{Variables:} \quad & p_{l,i}^{n,k}(t), \rho^{n,k}(t). \end{aligned}$$

Based on the optimal power control solution obtained from Problem (CMS-BOT-POW-C), the optimal channel allocation strategy can be found out by the CE algorithm similar to **Algorithm 2**.

C. MADRL for Bandwidth Assignment in Problem (TOP-Bandwidth)

In Problem (TOP-Bandwidth), each operator of the IS and CMS slices should request the bandwidth resources from infrastructure providers according to their own SLAs and users' QoS requirements. Due to the diverse SLAs and QoS requirements, it is challenging for conventional solutions to determine the optimal resource request, which aims at avoiding resource under-provisioning or over-provisioning while maximizing the spectral efficiency. The high mobility leads to the dynamic number of users, rapidly changing locations and per-user resource requirements, which also increases the uncertainty of resource request and the difficulty in resource assignment. Furthermore, centralized DRL algorithms require to collect the global information to optimize the action policy, which is inconsistent with the isolation and privacy of slices. Therefore, in this work, we propose a MADRL algorithm based on PPO algorithm, in which each agent only obtains its own partial observation space without requiring the global information to dynamically adjust its own bandwidth requests.

In this work, we use PPO algorithm, which is a well-known actor-critic policy-gradient algorithm[46], as our single basic agent algorithm. First, we formulate Problem (TOP-Bandwidth) as a Markov Decision Process, which comprises of state space $\mathcal{S}$, action space $\mathcal{A}$, reward function R and discount factor $\gamma \in [0,1]$. In particular, the state space, action space and reward function are defined as follows.

1) State Space: For the state space, the principle is to involve as much environment information in terms of Problem (TOP-Bandwidth) as possible. In Problem (TOP-Bandwidth), each slice as an independent agent has an independent observation space without accessing the global information. In the considering V2X communications system, the state space involves two kinds of slices including the number of users $I_m^{s,k}$ and $I_n^{c,k}$, the QoS requirements of users and the average queue length in last epoch $Q^{m,k}(0)$ or $Q^{n,k}(0)$. To keep the applicability of the state space for different IS and CMS slices, the QoS requirements of users consider three factors in both IS and CMS slices including the data rate requirement $r_m^{s,\min}$ or $r_n^{c,\min}$, delay requirement $P_m^{s,\text{d-max}}$ or $P_n^{c,\text{d-max}}$, and reliability requirement $P_m^{s,\text{r-max}}$ or $P_n^{c,\text{r-max}}$. Because there is no data rate constraint for CMS slices, $r_n^{c,\min}$ can be set to zero. Similarly, in IS slices, the reliability constraint $P_m^{s,\text{r-max}}$ can be set to 1. Besides, to achieve better convergence and coordination while protecting data privacy, slices cannot directly exchange information about their owns services but share the reward information at last epoch $\{R^{k-1}\}$. To this end, the state spaces of both the IS slice and CMS have same dimensions, and can be defined as follows

$$\mathcal{S}_m = \left\{ I_m^{s,k}, r_m^{s,\min}, P_m^{s,\text{d-max}}, P_m^{s,\text{r-max}}, Q^{m,k}(0), \left\{R^{k-1}\right\}\right\}, \tag{42}$$

$$\mathcal{S}_n = \left\{ I_n^{c,k}, r_n^{c,\min}, P_n^{c,\text{d-max}}, P_n^{c,\text{r-max}}, Q^{n,k}(0), \left\{R^{k-1}\right\}\right\}. \tag{43}$$

2) Action Space: In Problem (TOP-Bandwidth), the agent aims to determine the optimal bandwidth allocation policy. Thus the action space is the bandwidth assignment variables $W_m^s(k)$ and $W_n^c(k)$. As a result, the action space is given by

$$\mathcal{A}_m = \left\{ W_m^s(k) \right\}, \tag{44}$$

$$\mathcal{A}_n = \left\{ W_n^c(k) \right\}. \tag{45}$$

Due to that the applications served by CMS slices have stringent requirements on delay and reliability, it is important to provide them with adequate bandwidth resources. To provide better SLA satisfaction, the expert demonstration is employed to guarantee the minimum bandwidth assignment for CMS slices. According to [19], for each vehicle in CMS slices, the

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

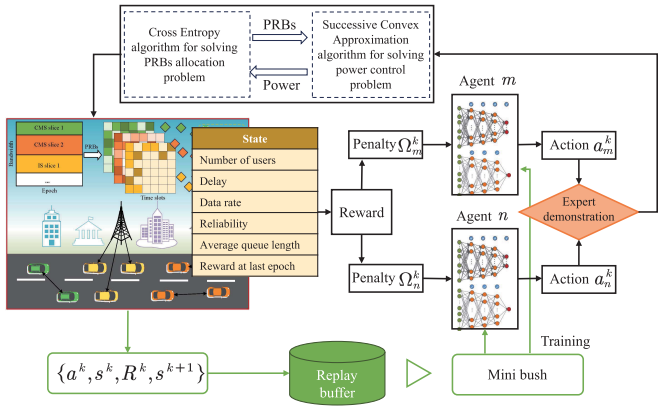


Fig. 2. The framework of our proposed MADRL algorithm.

required bandwidth to meet the minimum transmission rate with finite blocklength regime can be approximated as

$$B_n^{c,\mathrm{req}}(k) = \frac{r^{n,k-1,\mathrm{req}}}{\overline{C^{n,k-1}}} + \frac{\left(Q^{-1}(\varepsilon)\right)^2}{2\left(\overline{C^{n,k-1}}\right)^2} + \frac{\left(Q^{-1}(\varepsilon)\right)^2}{2\left(\overline{C^{n,k-1}}\right)^2}\sqrt{1+\frac{4r^{n,k-1,\mathrm{req}}\,\overline{C^{n,k-1}}}{\left(Q^{-1}(\varepsilon)\right)^2}}, \quad (46)$$

where $\overline{C^{n,k-1}} = \log_2(1+\overline{SNR^{n,k-1}})$ with $\overline{SNR^{n,k-1}}$ denoting the average SNR of vehicles in n-th CMS slice at the $(k-1)$-th epoch.

Then, the minimum bandwidth assigned to the n-th CMS slice at the k-th epoch according to the expert demonstration can be calculated as

$$W_n^{\exp}(k) = I_n^{c,k} B_n^{c,\mathrm{req}}(k). \quad (47)$$

If the action from the agent $W_n^c(k) \in \mathcal{A}_n$ is less than $W_n^{\exp}(k)$, we choose the expert policy $W_n^{\exp}(k)$ as the final action. Otherwise the $W_n^c(k) \in \mathcal{A}_n$ is set as the action.

3) Reward Function: The reward function needs to take into account both the objective function and the QoS constraints. To enable the agents to cooperatively maximize the objective function, the instant reward of each agent should include the same objective function $\mu\overline{SE} + \overline{EE}$. Meanwhile, each agent needs to compete for bandwidth resources to meet its users' QoS requirements. Thus, each agent should own independent penalty term to guarantee the constraints of QoS requirements. In IS slices, the agent should meet the data rate and delay requirements, and the penalty of IS slices can be defined as follows

$$\Omega_m = \varsigma^{\mathrm{pr}} \sum_{i\in\mathcal{I}_m^{s,k}} \aleph_{i,\mathrm{d}}^{m,k} + \sum_{i\in\mathcal{I}_m^{s,k}} \sum_{t\in\mathcal{T}} \aleph_{i,\mathrm{dr}}^{m,k}(t), \quad (48)$$

where

$$\aleph_{i,\mathrm{dr}}^{m,k}(t) = \begin{cases} 1, & r_i^{m,k}(t) - r_m^{s,\min} < 0, \\ 0, & r_i^{m,k}(t) - r_m^{s,\min} \geq 0, \end{cases} \quad (49)$$

and

$$\aleph_{i,\mathrm{d}}^{m,k} = \begin{cases} 1, & P_m^{s,\mathrm{d\text{-}max}} - \Pr\left\{Q_i^{m,k}(t) \geq Q_m^{s,\max}\right\} < 0, \\ 0, & P_m^{s,\mathrm{d\text{-}max}} - \Pr\left\{Q_i^{m,k}(t) \geq Q_m^{s,\max}\right\} \geq 0. \end{cases} \quad (50)$$

Similarly, the agent of CMS slices needs to guarantee the delay and reliability requirements, the penalty of which is given by

$$\Omega_n = \sum_{i\in\mathcal{I}_n^{c,k}} \aleph_{i,\mathrm{d}}^{n,k} + \aleph_{i,\mathrm{r}}^{n,k}, \quad (51)$$

where

$$\aleph_{i,\mathrm{d}}^{m,k} = \begin{cases} 1, & P_n^{c,\mathrm{d\text{-}max}} - \Pr\left\{Q_i^{n,k}(t) \geq Q_n^{c,\max}\right\} < 0, \\ 0, & P_n^{c,\mathrm{d\text{-}max}} - \Pr\left\{Q_i^{n,k}(t) \geq Q_n^{c,\max}\right\} \geq 0, \end{cases} \quad (52)$$

and

$$\aleph_{i,\mathrm{d}}^{m,k} = \begin{cases} 1, & P_n^{c,\mathrm{r\text{-}max}} - \Pr\left\{\gamma_i^{n,k}(t) \leq \gamma_n^{c,\min}\right\} < 0, \\ 0, & P_n^{c,\mathrm{r\text{-}max}} - \Pr\left\{\gamma_i^{n,k}(t) \leq \gamma_n^{c,\min}\right\} \geq 0. \end{cases} \quad (53)$$

Therefore, to strike the balance between the resource efficiency and the QoS requirements satisfaction, the reward function can be defined as

$$R_m = \mu\overline{SE} + \overline{EE} - \varsigma^s \Omega_m, \quad (54)$$

$$R_n = \mu\overline{SE} + \overline{EE} - \varsigma^c \Omega_n, \quad (55)$$

where ς^s and ς^c are the weight factors. Note that when any constraint in Problem (TOP-Bandwidth) is violated, the reward will receive a certain penalty. Therefore, to obtain a positive reward, the agent is required to not only maximize the objective function but also to guarantee that all the constraints are satisfied.

With the above defined action space, state space, and reward function, the framework of the proposed PPO-based MADRL is illustrated in Figure 2. The detailed processes are presented as follows. The PPO agent trains two separated deep neural networks including an actor network and a critic network. The actor network is responsible for selecting actions from continuous or discrete action space with the action policy π_θ, which maps the mean and standard deviation vectors of a Multivariate Gaussian Distribution. The objective of the trained actor network involves the advantage function $A(s, a)$ and the probability ratio of the current policy and the old policy p_θ^k, which is defined as

$$L(\theta) = \mathbb{E}_\pi\left[\min\left\{p_\theta^k A(s,a), \mathrm{clip}\left(p_\theta^k, 1-\epsilon, 1+\epsilon\right)A(s,a)\right\}\right], \quad (56)$$

where

$$A(s,a) = R(s,a) = \gamma V_\pi\left(s^{k+1}\right) - V_\pi\left(s^k\right), \quad (57)$$

$$p_\theta^k = \frac{\pi_\theta(s,a|\theta)}{\pi_\theta\left(s,a|\theta^{\mathrm{old}}\right)}, \quad (58)$$

and ϵ denotes the hyper-parameter which is used to adjust the clipping range. The critic network aims to evaluate the action selected by the actor network, which updates its parameter ϕ by minimizing the loss function

$$L(\phi) = \mathbb{E}_\phi\left[\max\left\{\left(p_\phi^k C(s,a) - R\right)^2, \Theta_\phi^k\right\}\right], \quad (59)$$

where

$$\Theta_\phi^k = \left(\mathrm{clip}(p_\phi^k, 1-\epsilon, 1+\epsilon)C(s,a) - R\right)^2. \quad (60)$$

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

Algorithm 3 MADRL Algorithm to Solve Problem (TOP-Bandwidth)

Initialization: Initialize θ, the parameters for policy π and ϕ, using the Orthogonal Initialization.
while epoch $k \leq K$ **do**
  for all agents **do**
    Initialize the environment and obtain initial state s^k.
    Select action a^k from the actor network.
    Compare the actions from the agent and expert demonstration, and select the maximization value.
    The slice optimizes PRBs allocation and power control in Problem (IS-BOT) and (CMS-BOT).
    Obtain the reward R^k according to (54) and (55).
    Update the next state s^{k+1}.
    if the replay buffer is not full **then**
      Push the tuple $\left\{a^k, s^k, R^k, s^{k+1}\right\}$ into the replay buffer.
    else
      Randomly replace a sample with the tuple $\left\{a^k, s^k, R^k, s^{k+1}\right\}$.
      Sample a mini-batch from the replay buffer.
      Update the parameters of critic network according to (59).
      Update the parameters of actor network according to (56).
    end if
  end for
end while

With the PPO framework, the agent at the epoch *K* selects an action a^k according the current state s^k from the actor network with policy π. After interacting with the environment, a reward R^k is obtained according to equations (54) and (55) and the next state s^{k+1} is generated. The tuple $\{a^k, s^k, R^k, s^{k+1}\}$ is stored into replay buffer. When the buffer is full, a mini-batch is sampled from the buffer and is used to train the actor and critic networks according to equations (56) and (59). The detailed process of our proposed MADRL is summarized in **Algorithm** 3.

D. Complexity Analysis

Problem (SEEE) is decomposed into a top-problem and a bottom-problem. In the top-problem, the proposed PPO-based agent is employed to output the actions according to the received state space. According to [47], the time complexity of our proposed agent with fully connected layers can be expressed as $O(\sum_{f=1}^{F} N_f^{\mathrm{cl}} N_{f-1}^{\mathrm{cl}})$, where N_f^{cl} denotes the number of neural units of the f-th fully connect layer. In the bottom-problem, we propose the CESCA algorithm to transform it into a convex optimization problem for optimizing the PRBs allocation and power control iteratively. The time complexity for solving the convex optimization problem is $O(N^{\max}\log(N^{\max}))$, where $N^{\max} = \max\{I_m^{s,k}, I_n^{c,k}\}_{\forall m\in\mathcal{M},\forall n\in\mathcal{N},\forall k\in\mathcal{K}}$. We assume that the iteration number of our proposed CESCA algorithm is N^{ite}. The maximum value of the time complexity for solving the bottom-problem with our proposed CESCA algorithm can be expressed as $O(N^{\mathrm{ite}}N^{\max}\log(N^{\max}))$. To this end, the overall complexity of our algorithms can be given by $O(\sum_{f=1}^{F} N_f^{\mathrm{cl}} N_{f-1}^{\mathrm{cl}} + N^{\mathrm{ite}}N^{\max}\log(N^{\max}))$.

TABLE II
SIMULATION PARAMETERS

Symbol	Value	Symbol	Value
W	20 MHz	$r_m^{s,\min}$	2 Mbps
Z	15 kHz	$P_m^{s,\text{d-max}}$	0.01
$P_n^{c,\text{d-max}}$	0.01	$P_n^{c,\text{r-max}}$	0.001
$Q_m^{s,\max}$	200 bits	$Q_n^{c,\max}$	200bits
μ	100	ϵ^{thr}	0.05
ς^s	20	ς^c	30
ς^{pr}	3	τ	1 ms
$p_m^{s,\max}$	1 W	$p_n^{c,\max}$	1 W
σ^2	-174 dBm/Hz	ϵ	0.8

V. SIMULATION AND PERFORMANCE EVALUATION

This section presents the simulations to evaluate the performance of our proposed MADRL algorithm. In this section, we first introduce the simulation settings and parameters, and then we introduce the several baseline approaches for comparison.

A. Network Architecture and Simulation Settings

The proposed algorithms are developed in a Python 3.8 environment with Pytorch 2.1.0. We perform the training and evaluation on an NVIDIA GeForce RTX 2080Ti GPU. The PPO-based agent includes the actor and critic networks, which both have three hidden layers comprising of 128, 128 and 64 neural units, respectively. The activation functions of the actor network are the Tanh function, which outperforms the ReLU function in the PPO algorithm, while the output layer applies the Sigmoid function to bound the bandwidth within the maximum value. The critic network uses the ReLU function as the activation functions of hidden layers. The learning rates of the actor and critic networks are set as 1×10^{-4} and 2×10^{-4}, respectively, which are the best settings we found during the testing. Considering the effect of the high mobility of vehicles in practical scenarios, we utilize the road traffic simulation software SUMO to construct the simulation environment according to the real-world topologies. The traffic data from the SUMO is then used for our simulations. The parameters related to the system model are listed in Table II.

B. Baseline Approaches

Our proposed algorithms consist of two parts, i.e., MAPPO algorithm with the expert demonstration mechanism for the inter-slice bandwidth allocation and CESCA algorithm for the intra-slice PRBs allocation and power control. To fully evaluate the performances of our proposed algorithm, we form the following three baseline approaches:

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

- Multi-agent deep determined policy gradient (MADDPG) [37] bandwidth allocation and CESCA PRBs allocation and power control (MAD-CESCA): In this approach, each slice acts as an independent agent to determine the bandwidth request from the well-trained MADDPG agent, and then obtains the optimal PRBs allocation and power control strategy according to the CE and SCA algorithms.
- MAPPO bandwidth allocation and stochastic PRBs allocation and fixed power control (MAP-SAFPC): In this approach, each slice acts as an independent agent to determine the bandwidth request from well-trained MAPPO agent. Then slices randomly allocate the PRBs to vehicles and a fixed transmit power.
- MADDPG bandwidth allocation and stochastic PRBs allocation and fixed power control (MAD-SAFPC): In this approach, each slice acts as an independent agent to determine the bandwidth request from well-trained MADDPG agent. Then slices randomly allocate the PRBs to vehicles and a fixed transmit power.

C. Results and Analysis

We first illustrate the performance of our proposed algorithm in terms of system spectral efficiency in comparison with the three baseline approaches. As shown in Figure 3, the system spectral efficiency increases monotonously as the number of vehicles increases for all the simulated approaches. This is because that more vehicles with shared bandwidth resources can take full advantage of the bandwidth resources and improve the bandwidth utilization. Moreover, due to the fewer bandwidth resources, vehicles need to increase the transmit power to guarantee the vehicles' QoS requirements, which leads to a higher spectral efficiency but a decreased energy efficiency. When the number of vehicles is more than 9, the performance gains on the spectral efficiency become smaller. This is because that the bigger and more complex inter-user interference results in a decreased transmission rate, which consequently leads to a reduction of performance gains on the spectral efficiency. Our proposed MAPPO algorithm with the CESCA algorithm outperforms other baseline approaches with different numbers of vehicles. This illustrates the cooperation between multiple agents in our algorithm works, which can maintain the optimal spectral efficiency without the global information. Furthermore, it can also be seen that our proposed CESCA scheme can improve the bandwidth utilization more than the SAFPC scheme when using the same algorithm for bandwidth assignment.

Figure 4 illustrates the performance in terms of the objective maximization of our proposed algorithm in comparison with the three baseline approaches with different numbers of vehicles. As shown in Figure 4, our proposed algorithm achieves the maximum objective with all different numbers of vehicles, and the proposed CESCA algorithm outperforms the SAFPC in terms of maximizing the spectral efficiency and energy efficiency. As the number of vehicles increases, the weighted sum of spectral efficiency and energy efficiency increases, but there is a smooth reduction in growing speed when the number

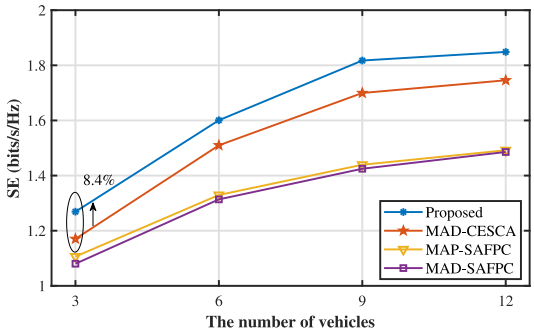


Fig. 3. The spectral efficiency with different numbers of vehicles.

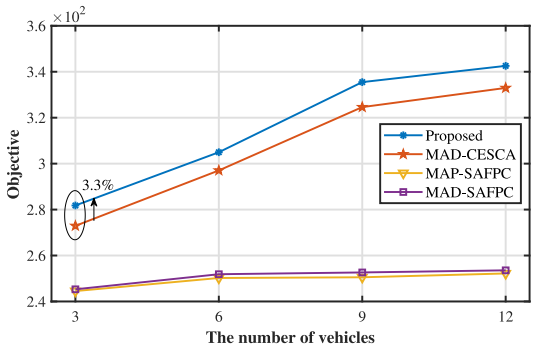


Fig. 4. The objective of maximizing the weighted sum of spectral efficiency and energy efficiency with different numbers of vehicles.

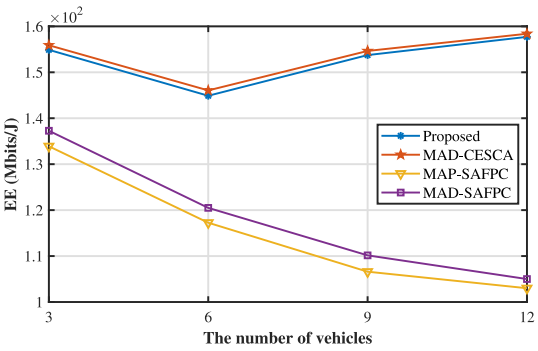


Fig. 5. The energy efficiency with different numbers of vehicles.

of vehicles is 6. The reason is that with the increase of the number of vehicles, the same bandwidth resources are not able to meet the vehicles' QoS requirements, so that more transmit power is required to increase the transmission rate and the energy efficiency thus decreases. Figure 5 also demonstrates this reason, in which the energy efficiency of all the simulated algorithms with the CESCA scheme reduces sharply when the number of vehicles is 6. It validates that with less spectrum resources, vehicles need to increase their transmit power at the expenses of the energy efficiency to guarantee the QoS requirements.

To further demonstrate the performance of our proposed MADRL algorithm, we use the centralized PPO algorithm [31] as the optimal algorithm for comparison. Figure 6 illustrates the accumulative reward per epoch during the MAPPO training process while allocating PRBs and optimizing power control with the CESCA scheme, compared with the centralized PPO algorithm. At the first 400th epochs, the agent explores the

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

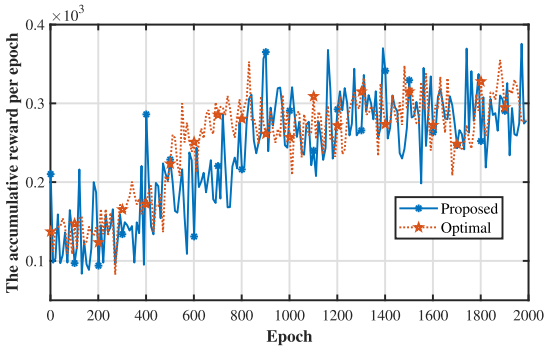


Fig. 6. The accumulative reward per epoch of our proposed algorithm and the optimal algorithm.

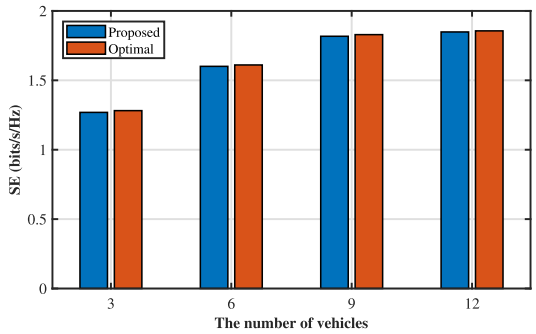


Fig. 7. The spectral efficiency of our proposed algorithm and the optimal algorithm with different numbers of vehicles.

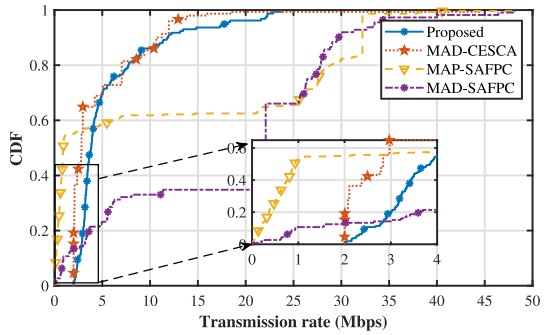


Fig. 8. The CDF of transmission rate in IS slices with different algorithms.

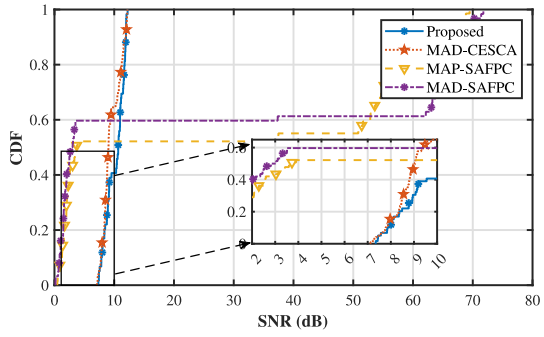


Fig. 9. The CDF of SNR in CMS slices with different algorithms.

state space as much as possible with a larger exploration noise. In this process, the agent aims to collect enough training samples while not updating the action policy. Therefore, the accumulative reward dynamically fluctuates. Then, the agent begins to learn and continuously updates its action policy to maximize the received reward. As the training goes on, the accumulative reward increases, and converges to the optimal value, i.e., around 300. Although the convergence speed of our proposed algorithm is slower than the centralized PPO algorithm with slight oscillation due to the competition between agents, our proposed algorithm is able to approximate to the centralized PPO algorithm, which has the best convergence performance due to making decisions with the global information. Figure 7 illustrates the performance of our proposed algorithm in terms of system spectral efficiency in comparison with the optimal algorithm. As shown in Figure 7, our proposed algorithm can achieve nearly optimal spectral efficiency without collecting the global information with all numbers of vehicles.

In order to demonstrate the performance of our proposed CESCA algorithm on guaranteeing the QoS requirements, the cumulative distribution functions (CDF) of the transmission rate in IS slices and the SNR in CMS slices are plotted in Figure 8 and Figure 9, respectively. The simulations consider all numbers of vehicles from 3 to 12. Figure 8 demonstrates the performance of our proposed CESCA algorithm on guaranteeing the minimum transmission rate requirement of IS slices. In IS slices, the required minimum transmission rate is set as 2 Mbps. As shown in Figure 8, the CDFs of the two CESCA-based algorithms, i.e., the proposed algorithm and the MAD-CESCA algorithm, are almost 0 when the transmission rate is no more than 2 Mbps. This demonstrates that the proposed CESCA-based algorithms can guarantee the minimum transmission rate requirement in IS slices effectively. On the contrary, the MAD-SAFPC algorithm and MAP-SAFPC algorithm fail to meet the requirements, the CDFs of which are about 10% and 55%, respectively. Figure 9 illustrates that the minimum SNR with the proposed CESCA algorithm is more than the required 7 dB, which guarantee the reliability requirement in CMS slices effectively. In conclusion, the proposed CESCA algorithm can effectively guarantee the QoS requirements in both IS and CMS slices.

VI. Conclusion

In this paper, we proposed a MAPPO algorithm and a CESCA algorithm to enable inter-slice bandwidth resources allocation and to optimize the intra-slice PRBs allocation and power control with the objective of maximizing the weighted sum of spectral efficiency and energy efficiency. In the long timescale, the PPO-based agent for each slice outputs the optimal acquired bandwidth resources according to the current partial environment to maximizing the spectral efficiency while meeting its SLA in terms of IS and CMS slices. In the short-term time slot, each slice allocates its available PRBs with the CE algorithm and then optimizes the power control with the SCA algorithm for the energy efficiency maximization while guaranteeing the transmission rate, delay and reliability requirements. Simulation results have demonstrated that our proposed MAPPO algorithm achieves approximate performance compared to the centralized PPO algorithm with the global information. Besides, our proposed

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

algorithm outperforms other baseline approaches in terms of the weighted sum of spectral efficiency and energy efficiency while guaranteeing the vehicles' QoS requirements.

REFERENCES

[1] W. Gao, C. Wu, L. Zhong, and K.-L. A. Yau, "Communication resources management based on spectrum sensing for vehicle platooning," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 2, pp. 2251–2264, Feb. 2023.

[2] M. H. C. Garcia et al., "A tutorial on 5G NR V2X communications," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1972–2026, 3rd Quart., 2021.

[3] G. Cai, B. Fan, Y. Dong, T. Li, Y. Wu, and Y. Zhang, "Task-efficiency oriented V2X communications: Digital twin meets mobile edge computing," *IEEE Wireless Commun.*, vol. 31, no. 2, pp. 149–155, Apr. 2024.

[4] W. Duan, J. Gu, M. Wen, G. Zhang, Y. Ji, and S. Mumtaz, "Emerging technologies for 5G-IoV networks: Applications, trends and opportunities," *IEEE Netw.*, vol. 34, no. 5, pp. 283–289, Oct. 2020.

[5] J. Wang, D. Li, H. Jiang, and M. Qiu, "Maximizing the connectivity of network slicing enabled Internet of Vehicle with differentiated services," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 13208–13217, Nov. 2023.

[6] S. Clayman et al., "The NECOS approach to end-to-end cloud-network slicing as a service," *IEEE Commun. Mag.*, vol. 59, no. 3, pp. 91–97, Mar. 2021.

[7] M. Richart, J. Baliosian, J. Serrat, and J. Gorricho, "Resource slicing in virtual wireless networks: A survey," *IEEE Trans. Netw. Service Manag.*, vol. 13, no. 3, pp. 462–476, Sep. 2016.

[8] Z. Chang, H. Deng, L. You, G. Min, S. Garg, and G. Kaddoum, "Trajectory design and resource allocation for multi-UAV networks: Deep reinforcement learning approaches," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 5, pp. 2940–2951, Oct. 2023.

[9] D. Sattar and A. Matrawy, "Optimal slice allocation in 5G core networks," *IEEE Netw. Lett.*, vol. 1, no. 2, pp. 48–51, Jun. 2019.

[10] G. Wang, G. Feng, T. Q. S. Quek, S. Qin, R. Wen, and W. Tan, "Reconfiguration in network slicing—Optimizing the profit and performance," *IEEE Trans. Netw. Service Manag.*, vol. 16, no. 2, pp. 591–605, Jun. 2019.

[11] J. J. Alves Esteves, A. Boubendir, F. Guillemin, and P. Sens, "A heuristically assisted deep reinforcement learning approach for network slice placement," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 4, pp. 4794–4806, Dec. 2022.

[12] R. Han et al., "Multi-SP network slicing parallel relieving edge network conflict," *IEEE Trans. Parallel Distrib. Syst.*, vol. 34, no. 11, pp. 2860–2875, Nov. 2023.

[13] C. Campolo, A. Molinaro, A. Iera, and F. Menichella, "5G network slicing for vehicle-to-everything services," *IEEE Wireless Commun.*, vol. 24, no. 6, pp. 38–45, Dec. 2017.

[14] Y. Chen, Y. Wang, M. Liu, J. Zhang, and L. Jiao, "Network slicing enabled resource management for service-oriented ultra-reliable and low-latency vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 7, pp. 7847–7862, Jul. 2020.

[15] S. Zhang, W. Quan, J. Li, W. Shi, P. Yang, and X. Shen, "Air-ground integrated vehicular network slicing with content pushing and caching," *IEEE J. Sel. Areas Commun.*, vol. 36, no. 9, pp. 2114–2127, Sep. 2018.

[16] A. Hossain and N. Ansari, "5G multi-band numerology-based TDD RAN slicing for throughput and latency sensitive services," *IEEE Trans. Mobile Comput.*, vol. 22, no. 3, pp. 1263–1274, Mar. 2023.

[17] M. K. Motalleb, V. Shah-Mansouri, S. Parsaeefard, and O. L. Alcaraz López, "Resource allocation in an open RAN system using network slicing," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 1, pp. 471–485, Mar. 2023.

[18] M. Setayesh, S. Bahrami, and V. W. Wong, "Joint PRB and power allocation for slicing eMBB and URLLC services in 5G C-RAN," in *Proc. IEEE Global Commun. Conf.*, Taipei, Taiwan, 2020, pp. 1–6.

[19] P. Yang, X. Xi, T. Q. S. Quek, J. Chen, X. Cao, and D. Wu, "How should I orchestrate resources of my slices for Bursty URLLC service provision?" *IEEE Trans. Commun.*, vol. 69, no. 2, pp. 1134–1146, Feb. 2021.

[20] M. Alsenwi, N. H. Tran, M. Bennis, S. R. Pandey, A. K. Bairagi, and C. S. Hong, "Intelligent resource slicing for eMBB and URLLC coexistence in 5G and beyond: A deep reinforcement learning based approach," *IEEE Trans. Wireless Commun.*, vol. 20, no. 7, pp. 4585–4600, Jul. 2021.

[21] G. Wang, G. Feng, W. Tan, S. Qin, R. Wen, and S. Sun, "Resource allocation for network slices in 5G with network resource pricing," in *Proc. IEEE Global Commun. Conf.*, Singapore, 2017, pp. 1–6.

[22] A. Fendt, C. Mannweiler, L. C. Schmelz, and B. Bauer, "An efficient model for mobile network slice embedding under resource uncertainty," in *Proc. 16th Int. Symp. Wireless Commun. Syst. (ISWCS)*, Oulu, Finland, 2019, pp. 602–606.

[23] Q. T. Luu, S. Kerboeuf, and M. Kieffer, "Admission control and resource reservation for Prioritized slice requests with guaranteed SLA under uncertainties," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 3, pp. 3136–3153, Sep. 2022.

[24] G. Dandachi, A. De Domenico, D. T. Hoang, and D. Niyato, "An artificial intelligence framework for slice deployment and orchestration in 5G networks," *IEEE Trans. Cogn. Commun. Netw.*, vol. 6, no. 2, pp. 858–871, Jun. 2020.

[25] J. S. Pujol Roig, D. M. Gutierrez-Estevez, and D. Gündüz, "Management and orchestration of virtual network functions via deep reinforcement learning," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 2, pp. 304–317, Feb. 2020.

[26] B. Han, V. Sciancalepore, X. Costa-Pérez, D. Feng, and H. D. Schotten, "Multiservice-based network slicing orchestration with impatient tenants," *IEEE Trans. Wireless Commun.*, vol. 19, no. 7, pp. 5010–5024, Jul. 2020.

[27] W. Jiang, Y. Zhan, G. Zeng, and J. Lu, "Probabilistic-forecasting-based admission control for network slicing in software-defined networks," *IEEE Internet Things J.*, vol. 9, no. 15, pp. 14030–14047, Aug. 2022.

[28] H. Peng and X. Shen, "Multi-agent reinforcement learning based resource management in MEC- and UAV-assisted vehicular networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 1, pp. 131–141, Jan. 2021.

[29] L. Niu et al., "Multiagent meta-reinforcement learning for Optimized task scheduling in heterogeneous edge computing systems," *IEEE Internet Things J.*, vol. 10, no. 12, pp. 10519–10531, Jun. 2023.

[30] B. Lu, B. Fan, Y. Wu, L. Qian, H. Zhang, and R. Lu, "Predictive computation offloading and resource allocation in DT-empowered vehicular networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 6, pp. 5474–5487, Jun. 2024.

[31] A. Awad Abdellatif, A. Abo-Eleneen, A. Mohamed, A. Erbad, N. V. Navkar, and M. Guizani, "Intelligent-slicing: An AI-assisted network slicing framework for 5G-and-beyond networks," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 2, pp. 1024–1039, Jun. 2023.

[32] J. J. Alcaraz, F. Losilla, A. Zanella, and M. Zorzi, "Model-based reinforcement learning with kernels for resource allocation in RAN slices," *IEEE Trans. Wireless Commun.*, vol. 22, no. 1, pp. 486–501, Jan. 2023.

[33] Y. Hua, R. Li, Z. Zhao, X. Chen, and H. Zhang, "GAN-powered deep distributional reinforcement learning for resource management in network slicing," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 2, pp. 334–349, Feb. 2020.

[34] J. Mei, X. Wang, and K. Zheng, "Semi-Decentralized network slicing for reliable V2V service provisioning: A model-free deep reinforcement learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 12108–12120, Aug. 2022.

[35] G. Sun, G. O. Boateng, D. Ayepah-Mensah, G. Liu, and J. Wei, "Autonomous resource slicing for virtualized vehicular networks with D2D communications based on deep reinforcement learning," *IEEE Syst. J.*, vol. 14, no. 4, pp. 4694–4705, Dec. 2020.

[36] G. O. Boateng, G. Sun, D. A. Mensah, D. M. Doe, R. Ou, and G. Liu, "Consortium blockchain-based spectrum trading for network slicing in 5G RAN: A multi-agent deep reinforcement learning approach," *IEEE Trans. Mobile Comput.*, vol. 22, no. 10, pp. 5801–5815, Oct. 2023.

[37] L. Tang, Y. Du, Q. Liu, J. Li, S. Li, and Q. Chen, "Digital-twin-assisted resource allocation for network slicing in industry 4.0 and beyond using distributed deep reinforcement learning," *IEEE Internet Things J.*, vol. 10, no. 19, pp. 16989–17006, Oct. 2023.

[38] C. Guo, C. Wang, L. Cui, Q. Zhou, and J. Li, "Radio resource management for C-V2X: From a hybrid Centralized-distributed scheme to a distributed scheme," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 4, pp. 1023–1034, Apr. 2023.

[39] C. Qi, Y. Hua, R. Li, Z. Zhao, and H. Zhang, "Deep reinforcement learning with discrete Normalized advantage functions for resource management in network slicing," *IEEE Commun. Lett.*, vol. 23, no. 8, pp. 1337–1341, Aug. 2019.

[40] C. Guo, L. Liang, and G. Y. Li, "Resource allocation for low-latency vehicular communications: An effective capacity perspective," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 4, pp. 905–917, Apr. 2019.

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.

[41] Y. Polyanskiy, H. V. Poor, and S. Verdu, "Channel coding rate in the finite blocklength regime," *IEEE Trans. Inf. Theory*, vol. 56, no. 5, pp. 2307–2359, May 2010.
[42] X. Zhang, M. Peng, S. Yan, and Y. Sun, "Deep-reinforcement-learning-based mode selection and resource allocation for cellular V2X communications," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6380–6391, Jul. 2020.
[43] T. K. Vu, C.-F. Liu, M. Bennis, M. Debbah, M. Latva-aho, and C. S. Hong, "Ultra-reliable and low latency communication in mmWave-enabled massive MIMO networks," *IEEE Commun. Lett.*, vol. 21, no. 9, pp. 2041–2044, Sep. 2017.
[44] L. Liang, G. Y. Li, and W. Xu, "Resource allocation for D2D-enabled vehicular communications," *IEEE Trans. Commun.*, vol. 65, no. 7, pp. 3186–3197, Jul. 2017.
[45] J. Tang, B. Shim, and T. Q. S. Quek, "Service multiplexing and revenue maximization in sliced C-RAN incorporated with URLLC and multicast eMBB," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 4, pp. 881–895, Apr. 2019.
[46] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.
[47] H. Gai, H. Zhang, S. Guo, and D. Yuan, "Information freshness-oriented trajectory planning and resource allocation for UAV-assisted vehicular networks," *China Commun.*, vol. 20, no. 5, pp. 244–262, May 2023.



Binbin Lu received the B.Sc. degree in logistics engineering and the M.S. degree in control science and engineering from Shandong University, Jinan, China, in 2019 and 2022, respectively. He is currently pursuing the Ph.D. degree with the Department of Computer and Information Science, University of Macau, Macau, China. His research interests include Internet of Vehicles, digital twin, and deep reinforcement learning.



Yuan Wu (Senior Member, IEEE) received the Ph.D. degree in electronic and computer engineering from the Hong Kong University of Science and Technology in 2010. He is currently an Associate Professor with the State Key Laboratory of Internet of Things for Smart City, Department of Computer and Information Science, University of Macau, Macau, China. His research interests include resource management for wireless networks, edge computing and edge intelligence, and integrated sensing and communication. He received the Best Paper Award from the IEEE ICC'2016, IEEE TCGCC'2017, IWCMC'2021, and WCNC'2023. He served as the Track/Symposium Co-Chair for IEEE VTC'2017-Fall, VTC'2021-Spring, VTC'2022-Spring, ICCC'2023, and GLOBECOM'2024. He is currently an Editorial Board of the IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, the IEEE TRANSACTIONS ON NETWORK SCIENCE AND ENGINEERING, and IEEE INTERNET OF THINGS JOURNAL.



Liping Qian (Senior Member, IEEE) received the Ph.D. degree in information engineering from The Chinese University of Hong Kong in 2010, where she worked as a Postdoctoral Research Associate from 2010 to 2011. Since 2011, she has been with the College of Information Engineering, Zhejiang University of Technology, Hangzhou, China, where she is currently a Full Professor. From 2016 to 2017, she was a Visiting Scholar with the Broadband Communications Research Group, Electronics and Communication Engineering Department, University of Waterloo. Her research interests include wireless communication and networking, resource management in wireless networks, massive IoTs, mobile edge computing, emerging multiple access techniques, and machine learning oriented towards wireless communications. She was a co-recipient of the IEEE Marconi Prize Paper Award in Wireless Communications in 2011, the Best Paper Award from IEEE ICC 2016, and the Best Paper Award from IEEE Communication Society GCCTC 2017. She is currently an Editorial Board of IET Communications.



Sheng Zhou (Senior Member, IEEE) received the B.E. and Ph.D. degrees in electronic engineering from Tsinghua University, Beijing, China, in 2005 and 2011, respectively, where he is currently an Associate Professor with the Department of Electronic Engineering. In 2010, he was a Visiting Student with the Wireless System Lab, Department of Electrical Engineering, Stanford University, Stanford, CA, USA. From 2014 to 2015, he was a Visiting Researcher with the Central Research Lab, Hitachi Ltd., Japan. His research interests include cross-layer design for multiple antenna systems, mobile edge computing, vehicular networks, and green wireless communications. He received the IEEE ComSoc Asia–Pacific Board Outstanding Young Researcher Award in 2017, and the IEEE ComSoc Wireless Communications Technical Committee Outstanding Young Researcher Award in 2020.



Haixia Zhang (Senior Member, IEEE) received the B.E. degree from the Department of Communication and Information Engineering, Guilin University of Electronic Technology, Guilin, China, in 2001, and the M.Eng. and Ph.D. degrees in communication and information systems from the School of Information Science and Engineering, Shandong University, Jinan, China, in 2004 and 2008, respectively, where she is currently a Distinguished Professor. From 2006 to 2008, she was an Academic Assistant with the Institute for Circuit and Signal Processing, Munich University of Technology, Munich, Germany. From 2016 to 2017, she was a Visiting Professor with the University of Florida, Gainesville, FL, USA. She is actively participating in many professional services. Her research interests include wireless communication and networks, industrial Internet of Things, wireless resource management, and mobile edge computing. She is an Editor of the IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE WIRELESS COMMUNICATION, and *China Communications* and a Symposium Chairs, the TPC Members, a Session Chairs, and the Keynote Speakers of many conferences.



Rongxing Lu (Fellow, IEEE) received the Ph.D. degree from the Department of Electrical and Computer Engineering, University of Waterloo, Canada, in 2012. He worked as a Postdoctoral Fellow with the University of Waterloo from May 2012 to April 2013. He is currently a Mastercard IoT Research Chair, a University Research Scholar, and a Professor with the Faculty of Computer Science, University of New Brunswick, Canada. Before that, he worked as an Assistant Professor with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, from April 2013 to August 2016. His research interests include applied cryptography, privacy enhancing technologies, and IoT-big data security, and privacy. He also serves as the Chair for IEEE Communications and Information Security Technical Committee, and the Founding Co-Chair for IEEE TEMS Blockchain and Distributed Ledgers Technologies Technical Committee.

Authorized licensed use limited to: Shaanxi Normal University. Downloaded on June 02,2026 at 03:15:09 UTC from IEEE Xplore. Restrictions apply.